Daughters Ahead, Sons Behind?

Economic Mobility and Educational Choices among Children of Immigrants in Scandinavia

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- Children of immigrants from comparable parental positions converge with native-origin peers

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1. Estimate immigrant-native earnings mobility in DK, SE, NO

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1. Estimate immigrant-native earnings mobility in DK, SE, NO
2. Detailed educational decompositions
3. Do **gendered patterns** of attainment $\Rightarrow$ gendered patterns of economic mobility?

Gendered mobility patterns?

Daughters

- Higher rates of educational completion
- Selection into high-return fields (law, medicine, business)
- Credentialed labor markets reward qualifications more reliably

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$\Rightarrow$ **Compounding disadvantage?**

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- Generous welfare provisions $\rightarrow$ dampen immigrants' work incentives?
- Strong welfare state institutions reduce negative consequences of childhood poverty
- Open educational systems $\rightarrow$ opportunities for immigrants' upward mobility

Data from 3 Scandinavian Countries

Sample: Full population, birth cohorts 1970–1989:

- All natives and descendants, observed at age 30–34
- **Children of immigrants:** born locally to two foreign-born parents, or arrived $<$ age 8
- Detailed info on parents, education, earnings

Result 1: Intergenerational convergence

(a) Sweden

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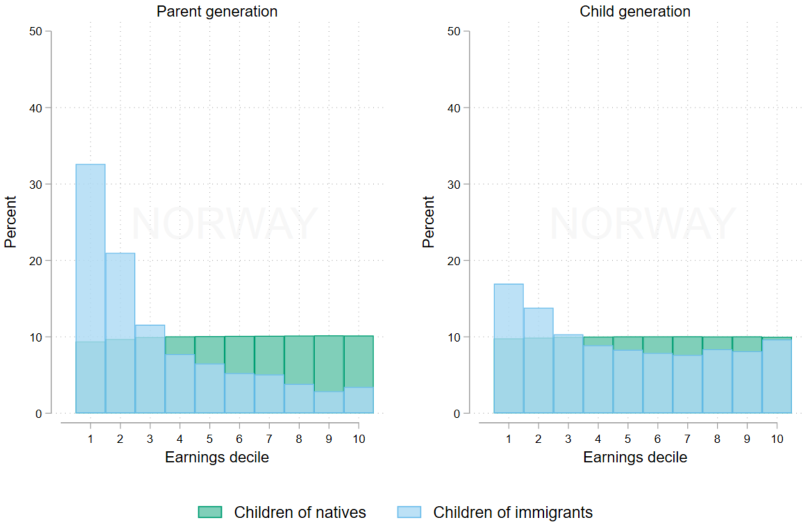
(b) Norway



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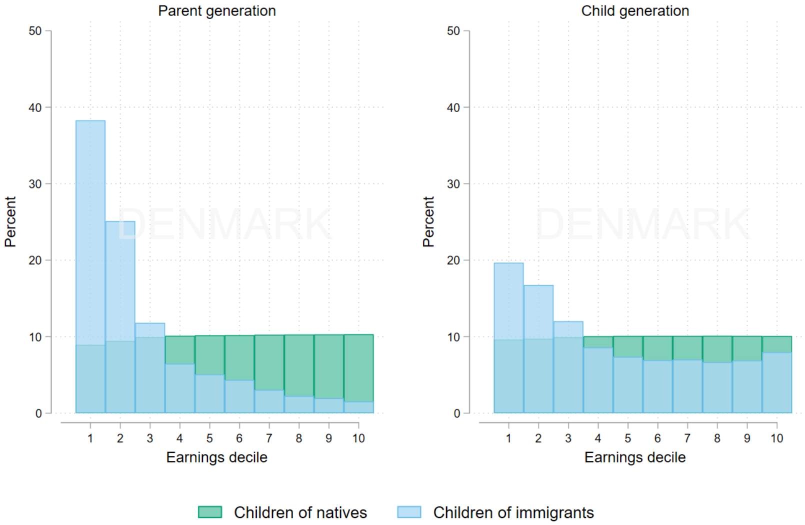

Baseline rank-rank regressions

Estimate separately for natives and immigrants $g \in \{N, I\}$:

$$y_i^C = \alpha_g + \beta_g y_i^P + \varepsilon_i$$

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Parent earnings rank at child age 13–20

Earnings: pre-tax income from wages, self-employment, capital gains

Ranks: within child birth cohorts, pooled across gender/ethnicity (1–100)

Result 2: rank-rank regressions

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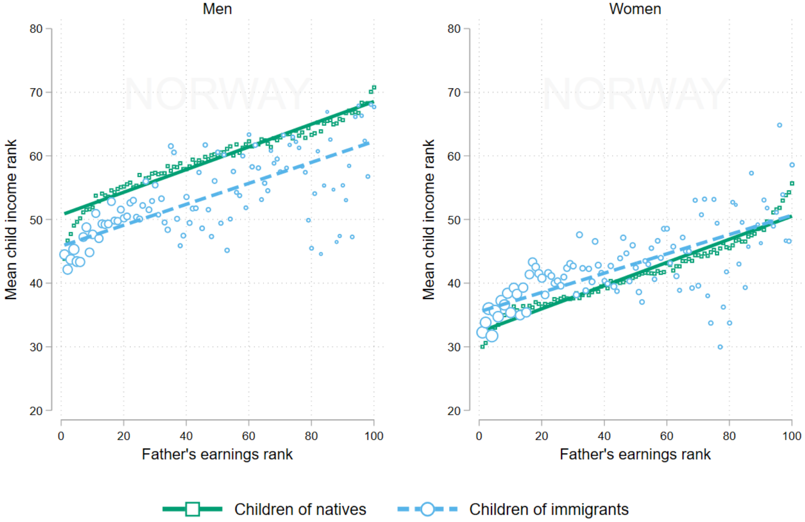

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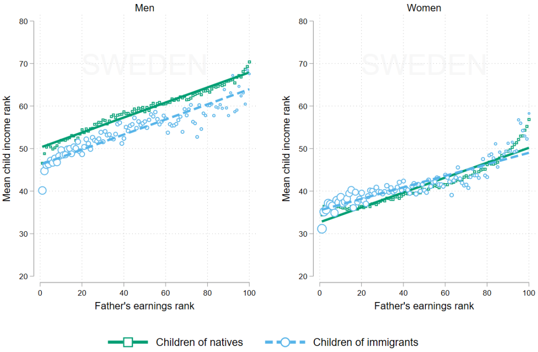

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- Large absolute upward mobility

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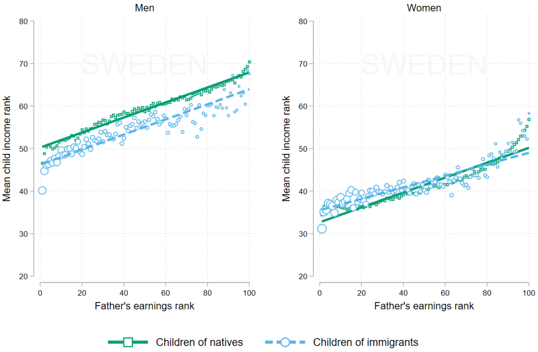

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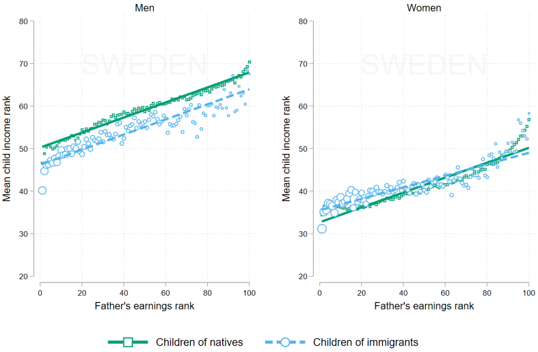

- Overall strikingly similar patterns across Scandinavia
- Large absolute upward mobility
- **Men:** immigrants **earn less** across the entire parental distribution (level gap — similar slope)
- **Women:** immigrants **match/exceed** natives (especially at lower parent ranks)

Educational decompositions

$$y_i^C = \alpha_g + \beta_g y_i^P + \delta_g \text{Educ}_i + \varepsilon_i \tag{1}$$

Result 3: Education explains gap for women

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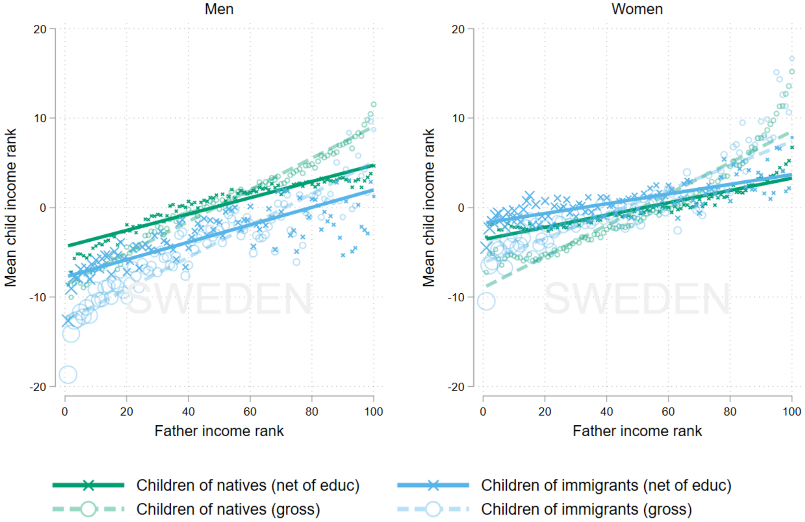

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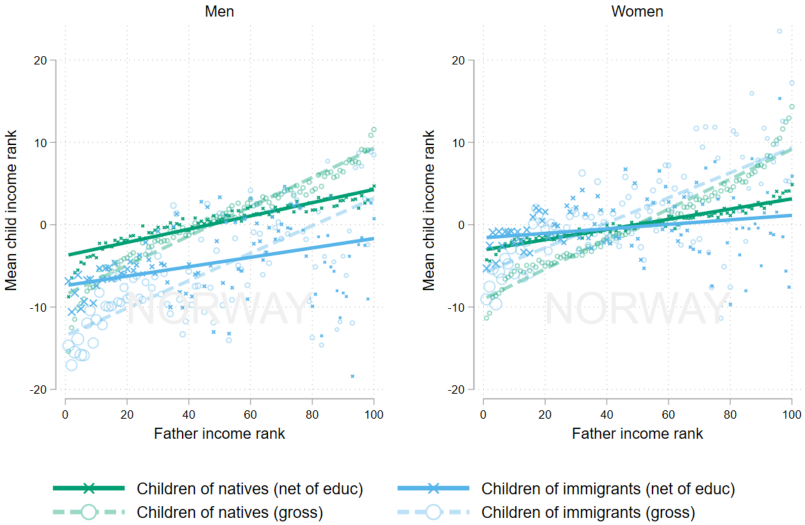

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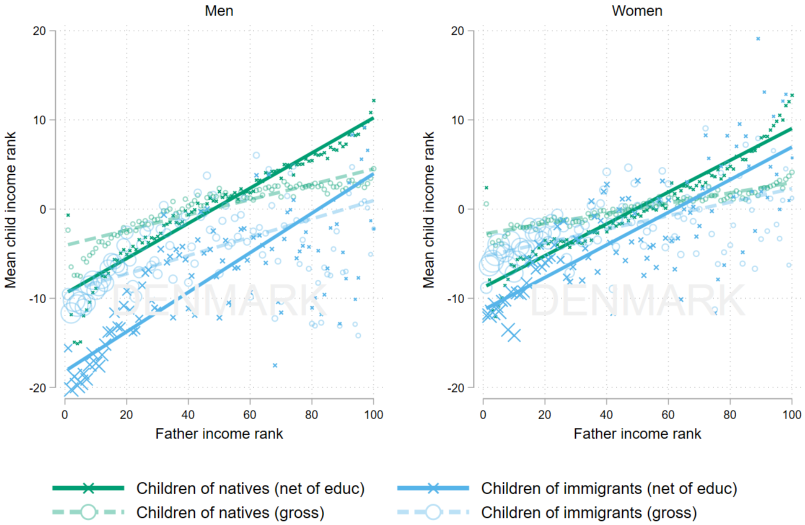

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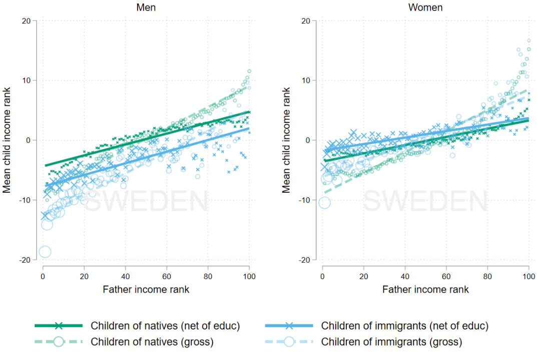

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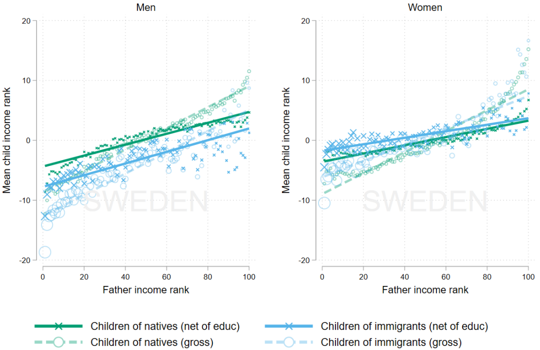

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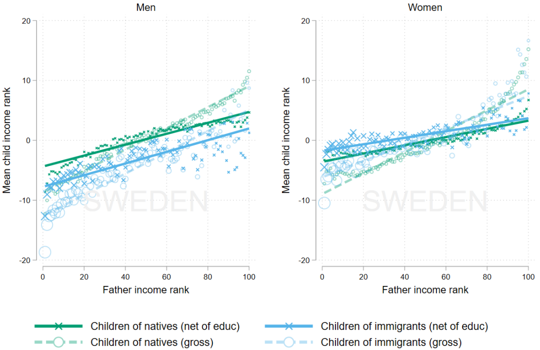

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⇒ Immigrant male disadvantage operating through *unequal returns*, rather than *attainment?*

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Standard decomposition:

$$\beta_I - \beta_N = \underbrace{(\beta_I^* - \beta_N^*)}_{\text{Within-education}} + \underbrace{\left[(\beta_I - \beta_I^*) - (\beta_N - \beta_N^*)\right]}_{\text{Between-education}}$$

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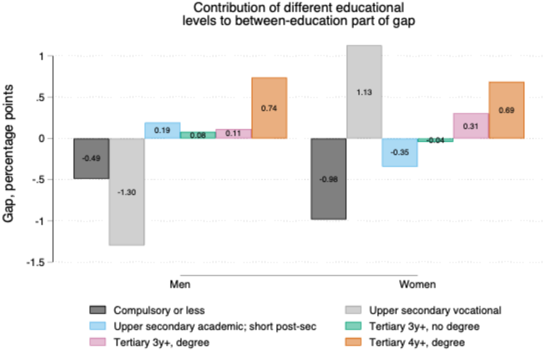
The between-education term further decomposes into:

$$\Delta\beta^{\text{between}} = \sum_e \underbrace{(\lambda_e^I - \lambda_e^N)\,\theta_e^N}_{\text{(a) attainment differences}} + \sum_e \underbrace{\lambda_e^I\,(\theta_e^I - \theta_e^N)}_{\text{(b) return differences}}$$

Derivation

Result 4: Educational decomposition — Attainment



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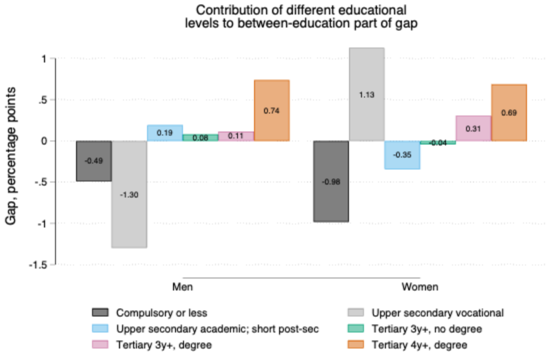

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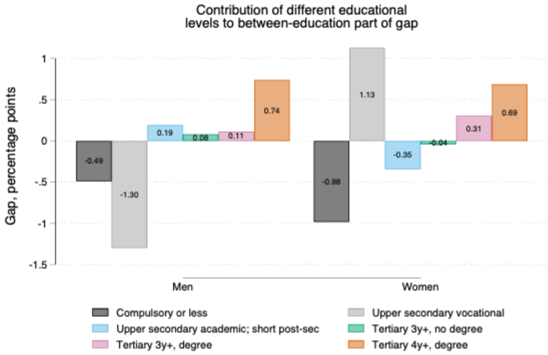

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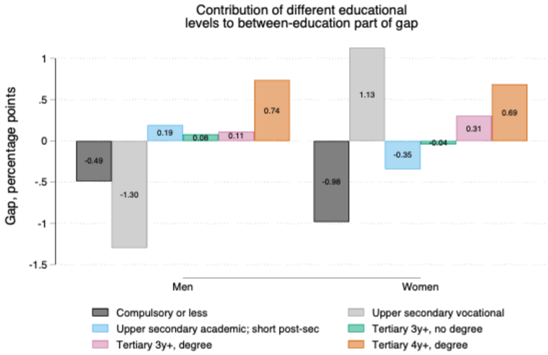

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- **Sons:** gap survives at every education level $\implies$ driven by *returns*, not attainment
- **Daughters:** advantage comes from *attainment* — reaching higher levels, especially in high-return fields

Conclusions

- Strikingly similar patterns of intergenerational earnings mobility among immigrants in Scandinavia
- Male **disadvantage** and female **advantage** in adult earnings, conditional on parental earnings
- Education explains the female advantage but not the male disadvantage
- Gender differences in educational aspirations versus persistent barriers in the labor market?

Thank you!

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Appendix: derivation of the between-education decomposition

Step 1. Estimate the rank–rank slope with and without education controls, separately by group $g \in \{I, N\}$:

$$y_i^C = \alpha_g + \beta_g\, y_i^P + \varepsilon_i$$
$$y_i^C = \alpha_g^* + \beta_g^*\, y_i^P + \delta_g\, \text{Educ}_i + \epsilon_i$$

Step 2. Auxiliary regressions identify the two channels:

$$\text{Educ}_{ie} = \kappa_e + \lambda_e^g\, y_i^P + u_{ie} \qquad (\lambda_e^g\text{: parent rank} \rightarrow \text{attaining level } e)$$
$$y_i^C = \textstyle\sum_e \theta_e^g\, \text{Educ}_{ie} + \varepsilon_i \qquad (\theta_e^g\text{: return to level } e\text{, net of } y^P)$$

Step 3. Total slope difference decomposes as:

$$\beta_I - \beta_N = \underbrace{(\beta_I^* - \beta_N^*)}_{\text{Within-education}} + \underbrace{\left[(\beta_I - \beta_I^*) - (\beta_N - \beta_N^*)\right]}_{\text{Between-education} = \Delta\beta^{\text{between}}}$$

Step 4. The between-education term separates the two mechanisms:

$$\Delta\beta^{\text{between}} = \sum_e \underbrace{(\lambda_e^I - \lambda_e^N)\,\theta_e^N}_{\text{(a) attainment differences}} + \sum_e \underbrace{\lambda_e^I\,(\theta_e^I - \theta_e^N)}_{\text{(b) return differences}}$$

Per-level contribution: $\Delta\beta_p^{(e)} = (\lambda_e^I - \lambda_e^N)\,\theta_e^N + \lambda_e^I\,(\theta_e^I - \theta_e^N)$, with $\Delta\beta^{\text{between}} = \sum_e \Delta\beta_p^{(e)}$.

References I

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